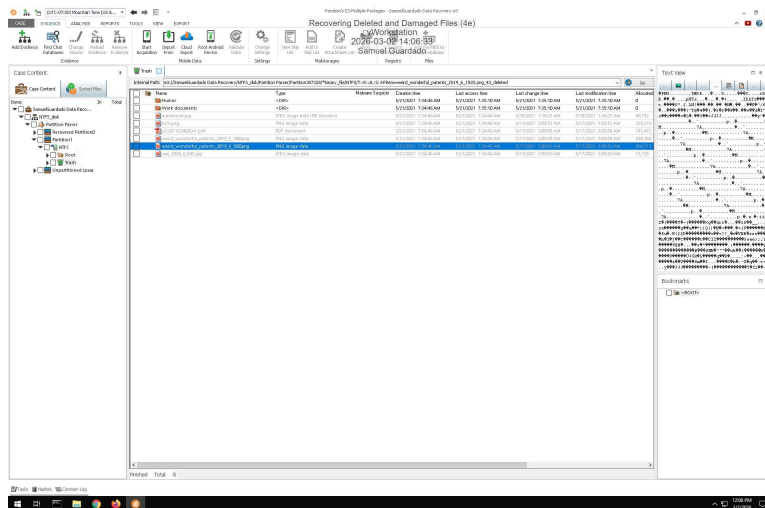


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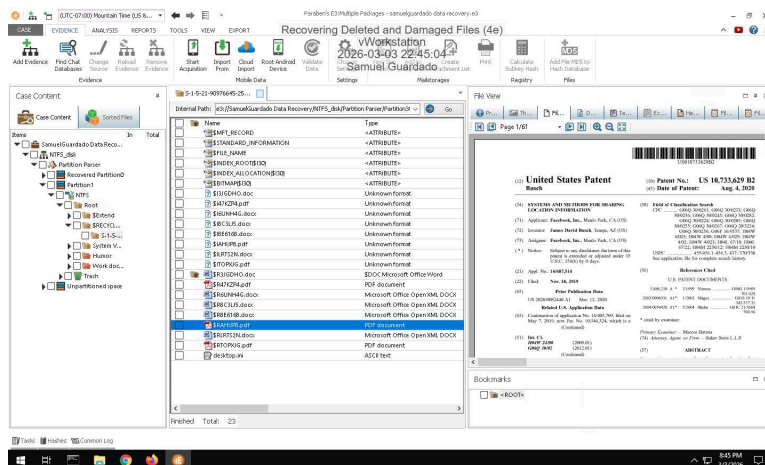
Time on Task: 3 hours, 42 minutes
Progress: 100%

Part 1: Recover Deleted Files from an NTFS Drive Image with E3

13. **Make a screen capture** showing the **list of recovered files and folders in the E3 Trash folder**.



20. **Make a screen capture** showing the patent file in the File Viewer.



25. **Make a screen capture** showing the **recovered files in the File Explorer**.


```
Activities Terminal Mar 6 23:16
Recovering Deleted and Damaged Files (4e)
TargetLinux01
2026-03-06 23:16:07
Samuel Guardado
u Update files with size=1000 [*1024, *1]
v<size>[k,b] Create volumes with size=1000 [*1024, *1]
ver[n] File version control
vn Use the old style volume naming scheme
vp Pause before each volume
w<path> Assign work directory
x<file> Exclude specified file
x@ Read file names to exclude from stdin
x@<list> Exclude files listed in specified list file
y Assume Yes on all queries
z[file] Read archive comment from file
user@TargetLinux01:/mnt/media/home/ash$ rar l backups.rar

RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5

Attributes Size Date Time Name
-----
-rw----- 0 2021-08-20 14:19 backups/.recoveryKeys.txt.swp
-rw----- 0 2021-08-20 14:20 backups/.recoveryKeys.txt.swo
-rw-r--r-- 20000000 2021-08-20 14:22 backups/serverImage.dd
-rw-r--r-- 257 2021-08-20 14:21 backups/recoveryKeys.txt
-----
20000257 4

user@TargetLinux01:/mnt/media/home/ash$
```

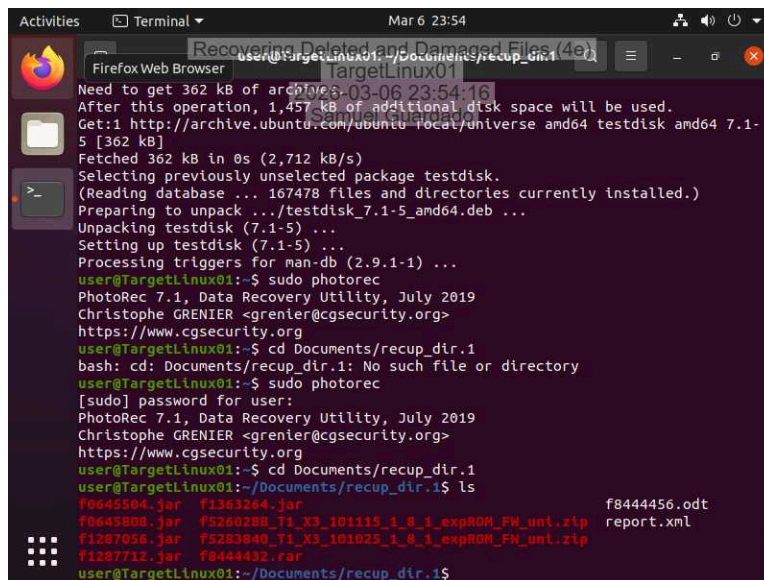
15. Make a screen capture showing the failed mount attempt on the /dev/sdb2 device.

```
Activities Terminal Mar 6 23:28
Recovering Deleted and Damaged Files (4e)
TargetLinux01
2026-03-06 23:28:25
Samuel Guardado
Archive: backups.rar
Details: RAR 5

Attributes Size Date Time Name
-----
-rw----- 0 2021-08-20 14:19 backups/.recoveryKeys.txt.swp
-rw----- 0 2021-08-20 14:20 backups/.recoveryKeys.txt.swo
-rw-r--r-- 20000000 2021-08-20 14:22 backups/serverImage.dd
-rw-r--r-- 257 2021-08-20 14:21 backups/recoveryKeys.txt
-----
20000257 4

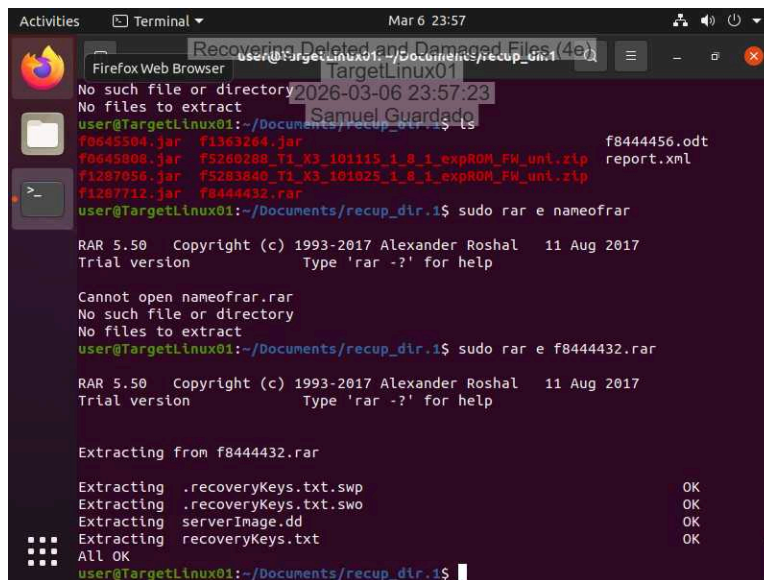
user@TargetLinux01:/mnt/media/home/ash$ rm -f *
user@TargetLinux01:/mnt/media/home/ash$ cd ~
user@TargetLinux01:~$ sudo umount /dev/sdb2
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k
4096+0 records in
4096+0 records out
4194304 bytes (4.2 MB, 4.0 MiB) copied, 0.246567 s, 17.0 MB/s
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, missing codepage or helper program, or other error.
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, missing codepage or helper program, or other error.
user@TargetLinux01:~$ sudo mount/dev/sdb2 /mnt/media
sudo: mount/dev/sdb2: command not found
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
```

32. Make a screen capture showing the compressed files recovered by PhotoRec.



```
Activities Terminal Mar 6 23:54
Recovering Deleted and Damaged Files (4e)
user@TargetLinux01:~/Documents/recup_dir.1$
Need to get 362 kB of archive:
After this operation, 1,457 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal/universe amd64 testdisk amd64 7.1-5 [362 kB]
Fetched 362 kB in 0s (2,712 kB/s)
Selecting previously unselected package testdisk.
(Reading database ... 167478 files and directories currently installed.)
Preparing to unpack .../testdisk_7.1-5_amd64.deb ...
Unpacking testdisk (7.1-5) ...
Setting up testdisk (7.1-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@TargetLinux01:~$ sudo photorec
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
user@TargetLinux01:~$ cd Documents/recup_dir.1
bash: cd: Documents/recup_dir.1: No such file or directory
user@TargetLinux01:~$ sudo photorec
[sudo] password for user:
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
user@TargetLinux01:~$ cd Documents/recup_dir.1
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645584.jar f1363244.jar f8444456.odt
f0645888.jar f5260288.T1_X3_101115_1_8_1_expROM_FW_uni.zip report.xml
f1287056.jar f5283840.T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287712.jar f8444432.rar
user@TargetLinux01:~/Documents/recup_dir.1$
```

35. Make a screen capture showing the backup files recovered from the RAR archive.

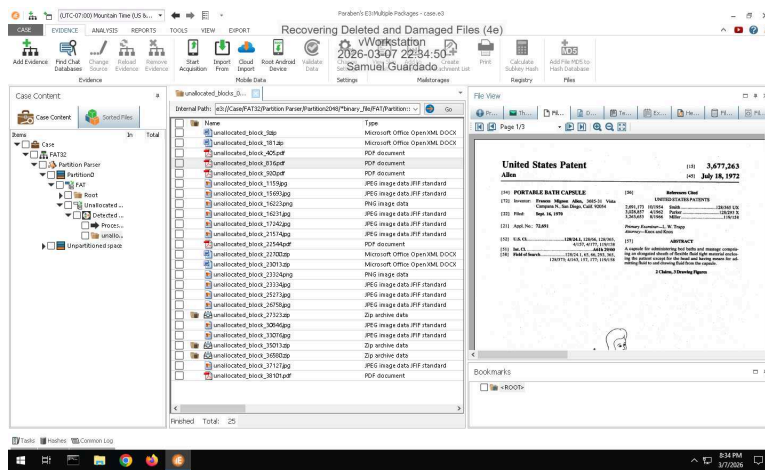


```
Activities Terminal Mar 6 23:57
Recovering Deleted and Damaged Files (4e)
user@TargetLinux01:~/Documents/recup_dir.1$
No such file or directory
No files to extract
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645584.jar f1363244.jar f8444456.odt
f0645888.jar f5260288.T1_X3_101115_1_8_1_expROM_FW_uni.zip report.xml
f1287056.jar f5283840.T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287712.jar f8444432.rar
user@TargetLinux01:~/Documents/recup_dir.1$ sudo rar e nameofrar
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help
Cannot open nameofrar.rar
No such file or directory
No files to extract
user@TargetLinux01:~/Documents/recup_dir.1$ sudo rar e f8444432.rar
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help
Extracting from f8444432.rar
Extracting .recoveryKeys.txt.swp OK
Extracting .recoveryKeys.txt.swo OK
Extracting serverImage.dd OK
Extracting recoveryKeys.txt OK
All OK
user@TargetLinux01:~/Documents/recup_dir.1$
```

Section 3: Challenge and Analysis

Part 1: Recover Deleted Files from a FAT Drive Image

Make a screen capture showing the patent file recovered from the FAT32 drive image within E3.



Part 2: Recover Deleted Files from a APFS Drive Image

Make a screen capture showing the patent file recovered from the APFS drive image within Autopsy.

